MixIndex: Mixed-Fidelity Vector Indexes with Traffic-Driven Backfill for Embedding-Model Migration

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Abstract. A document search system represents each document as a vector produced by an embedding model. Upgrading that model changes the vector space, so existing document vectors may no longer match newly encoded queries. Recomputing every vector can require substantial processing time and a second index. We propose MixIndex, which maps old vectors into the new space and gradually replaces them with vectors computed directly by the new model. Both types share one index. We refine the mapping within groups of similar documents, calibrate their search scores, and use query traffic to choose which documents to recompute first. Our analysis gives conditions under which mapping preserves the highest-scoring results and bounds the benefit of greedy scheduling for a fixed coverage objective. We describe an evaluation across four model pairs, five retrieval collections, and public query logs to examine search quality, migration progress, and cost. The numerical results in this draft are illustrative placeholders; experiments remain to be run. Our design makes model migration a gradual process in which useful new-model representations can enter service before the whole collection is recomputed.

1 Introduction

A document search system uses an *embedding model* to turn text into a vector of numbers. The model is trained so that a query and a relevant document receive similar vectors. An index stores the document vectors and helps find those closest to a query vector. Approximate nearest-neighbour search makes this lookup fast by examining only part of the index, at the cost of sometimes missing a closer vector. Retrieval-augmented systems use the retrieved documents as context for generating an answer.

Why a model upgrade disrupts search. Consider a search for "who built the Eiffel Tower?" in a collection containing documents about the tower, its height, and its designers. The old model places these documents in one vector space. A new model may represent their meanings differently, even if it uses the same number of coordinates. Comparing a new query vector directly with old document vectors can therefore produce the wrong ranking. *Re-embedding* means running the new

model on the original document text to obtain its new vector. A *full rebuild* does this for every document and builds an index from the new vectors.

Vector databases document migration procedures that write vectors from both models, re-embed the collection in the background, and switch to the new index when it is ready [23,35,3]. This process can require hours or days of graphics processing unit (GPU) time, depending on collection size, as well as storage for a second index [31]. We ask whether search can begin using the new model while this work is still in progress.

From a temporary map to gradual migration. A learned map can approximate the relationship between the two vector spaces using a sample of documents encoded by both models. Drift-Adapter [31] maps new queries into the old document space. It reports 95 to 99% Recall@10 relative to the new model: the fraction of the new model's ten highest-scoring documents recovered by the adapted search. This preserves the old index, but its documents never acquire the new model's representations. Hot-Refresh [41] instead gradually recomputes an image gallery, a process called *backfill*. Its ordering uses classifier uncertainty and compatible model training, which are not generally available for independently trained text embedding models.

We propose MixIndex, which maps old *document* vectors forward into the new space. We call these predictions *mapped vectors*; vectors computed from the original text by the new model are *re-embedded vectors*. Our index contains both, so we call it a *mixed-fidelity index*. Here, fidelity records how a vector was obtained. Every query uses the new model, and backfill replaces mapped vectors in place [1]. In our running example, we can recompute a frequently retrieved Eiffel Tower document while less frequently used documents retain mapped vectors.

Three problems arise during this migration. First, a single map may fit some topics better than others. We therefore learn corrections within regions of similar old vectors and update them using pairs produced by backfill. Second, mapped and re-embedded vectors may have systematically different search scores. We calibrate scores by fidelity so that a document's representation type does not alone determine its rank. Third, a limited GPU budget forces us to choose which documents to recompute. We use query frequencies and score gaps to prioritize documents that could preserve a relevant result for frequently asked queries.

Contributions and evidence. We present the mixed index, regional mapping, score calibration, and traffic-based scheduling mechanism in Section 3. Section 4 explains when bounded score errors preserve the exact top-k ranking and when backfill preserves at least one relevant result. For a fixed coverage objective, our greedy scheduler has the classical $(1 - 1/e)$ approximation guarantee [21]; e is the base of the natural logarithm, so the factor is about 0.632. This guarantee concerns the scheduling objective under its stated assumptions.

Our planned evaluation compares four open model pairs, five BEIR retrieval collections, and MS MARCO documents with ORCAS and AOL-IA query traffic against nine baselines (Sections 5 and 6). We organize it around quality before

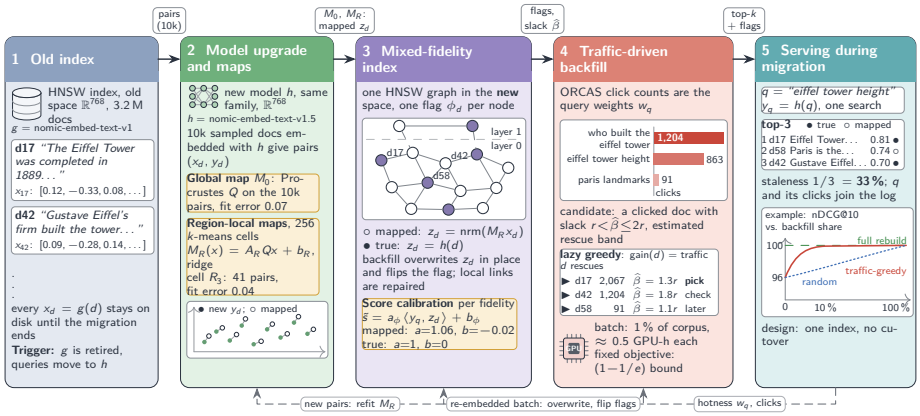


Fig. 1. An illustrative migration with MixIndex. We learn maps from documents encoded by both models, store mapped (○) and re-embedded (●) vectors together, and use query traffic to select backfill batches. The index supports hierarchical navigable small-world (HNSW) search. The highest-scoring k results are the top-k; score slack measures distance from their cutoff. The plotted ranking-quality measure, normalized discounted cumulative gain at rank 10 (nDCG@10), rewards relevant results near the top. The old and new models are g and h; x_d, y_d, and z_d are old, new, and stored document vectors, and y_q is a new query vector. The maps M_0 and M_R are global and regional (A_R, b_R are correction parameters); ϕ_d records fidelity. The error radius is r, query frequency is w_q, and a_ϕ, b_ϕ calibrate scores. The orthogonal map is Q, and nrm normalizes a vector to unit length. Values and curves illustrate the design and are not measurements.

backfill (**Q1**), quality as migration progresses (**Q2**), regional maps and calibration (**Q3**), and theory checks and cost (**Q4**). All numerical results in this draft are illustrative placeholders. They specify the intended comparisons and reporting format; they do not establish empirical performance.

2 Related Work

Keeping representations compatible. Some methods prepare models for future upgrades during training. Backward-compatible training [27] adds a loss that encourages new features to remain comparable with old ones. Forward-compatible training [24] stores additional information per item so that an old embedding can later be moved into a new space. Hot-Refresh [41] orders image-gallery backfill by classifier uncertainty. It also studies the negative flip rate: how often a previously correct result becomes incorrect after an update. We adopt ordered backfill and this stability measure, while assuming independently available model checkpoints and stored old vectors.

Learning a map between spaces. Drift-Adapter [31] maps new queries into the old document space. It uses orthogonal Procrustes [26], an alignment by rotation

or reflection; a low-rank affine map, a factored linear transform plus an offset; and a residual multilayer perceptron (MLP), a neural correction to a linear map. Query drift compensation (QDC) [8] uses a mean-shift vector to restore compatibility after continual fine-tuning. MURR [39] maintains multiple session-specific indexes instead of mapping between spaces.

Other methods change the information needed for alignment. vec2vec [11] and mini-vec2vec [6] align spaces without paired examples. Relative representations [19] describe each vector by its cosine similarities to a fixed set of anchor documents. Maystre et al. [16] analyze Procrustes alignment and report diminishing retrieval gains beyond roughly ten thousand fitting pairs. We use these approaches as starting points or baselines. Our design adds forward document mapping, regional updates from backfilled pairs, and score calibration between representation types.

Maintaining an index during migration. HNSW [15], DiskANN [28], and FAISS [7] provide vector-search indexes or libraries with update support. DiskANN targets search over indexes stored on disk. Embedding Surgery [1] studies overwriting vectors in HNSW and DiskANN without reconstructing the whole index, the operation on which our backfill design relies. Qdrant [23], Weaviate [35], and Vespa [3] document migration through concurrent writes and full background re-embedding.

A related line studies databases that already use different embedding models. FedBridge [33] and its underlying federation method [34] answer queries across such databases. Work on integrating vector databases across embedding models [38] addresses a related integration setting. Another approach reuses an old graph after recomputing every vector [9]. Our focus is the intermediate migration state, in which some vectors have been recomputed and others remain predictions in one serving index.

Evaluation resources and scheduling theory. BEIR [30] and the Massive Text Embedding Benchmark (MTEB) [20] evaluate retrieval and embedding quality. MS MARCO [2], ORCAS click logs [5], AOL-IA search logs with recovered documents [14], and TripClick [25] provide queries, relevance signals, or traffic information. Successive nomic-embed [22], Arctic-embed [17,40], bge [37,4], and e5 [32] models motivate migration across checkpoints.

Our analysis connects score error to ranking and then to backfill selection. Weller et al. [36] study limits on the top-k sets embeddings can represent. Nearest-neighbour-preserving embeddings [10] and the Johnson–Lindenstrauss lemma [12] provide context for distance preservation and random projection. For scheduling, a coverage function counts the weight of queries helped by at least one selected document. It is monotone and submodular: adding a document cannot reduce coverage, and its additional benefit decreases as other documents cover the same queries. Greedy selection has a $(1 - 1/e)$ guarantee under a document-count budget [21]. Lazy evaluation reduces repeated gain calculations [18,13]; partial enumeration supports the corresponding guarantee under

a budget with unequal costs [29]. Section 4 states which version applies to our design.

3 The MixIndex Mechanism

Setting. We use the document collection and query stream to decide how to spend a limited re-embedding budget. Let D be the corpus and $N = |D|$ its number of documents. For a document d, the old model g produces a unit vector $x_d = g(d) \in \mathbb{R}^{d_0}$. Here $\mathbb{R}^{d_0}$ is the space of real vectors with d_0 coordinates. The replacement model h produces $y_d = h(d) \in \mathbb{R}^{d_1}$, where $d_1 \geq d_0$. We normalize embeddings to unit length and encode each query q only with the new model, giving $y_q = h(q)$. Its weight w_q is its frequency in the query log.

The full rebuild computes every y_d and returns $T_k(q) = \mathrm{argtop}_k \langle y_q, y_d \rangle$. The inner product $\langle u, v \rangle$ sums coordinatewise products; for unit vectors it equals cosine similarity. The operator argtop_k selects the k document identifiers with the largest scores. Our budget permits re-embedding a fraction f of the corpus, or fN documents under equal per-document cost. We aim to keep the served ranking close to the full-rebuild ranking on arriving traffic while maintaining one serving index. Figure 1 traces the data flow.

3.1 Mixed-Fidelity Index

We use one hierarchical navigable small-world (HNSW) graph [15], an index whose links connect nearby vectors and guide approximate search. All stored vectors lie in the new space $\mathbb{R}^{d_1}$. Each document has a stored vector z_d and a fidelity flag $\phi_d \in \{\mathrm{mapped}, \mathrm{true}\}$. The flag `true` means that the new model has encoded the original document; it does not assert that the document is relevant. We store

$$z_d = \mathrm{nrm}\left(M_{R(d)}(x_d)\right) \text{ if } \phi_d = \mathrm{mapped}, \qquad z_d = y_d \text{ if } \phi_d = \mathrm{true}, \tag{1}$$

where $\mathrm{nrm}(v) = v/\|v\|_2$ divides a nonzero vector by its Euclidean length. The region assignment $R(d)$ identifies a group of similar old vectors, and M_R is the map for region R, defined below. Equation (1) thus distinguishes a normalized prediction from a directly computed embedding.

We build the index from mapped vectors. Each backfill batch overwrites selected vectors and changes their flags. For example, the Eiffel Tower article can receive its new embedding while its neighbouring articles remain mapped. Embedding Surgery [1] motivates reusing graph links after such replacements. Our design also repairs an overwritten node's neighbour list with one greedy search from its new vector. Section 6 specifies the planned check of this choice. Once all flags are true, the stored vectors match those of a full rebuild. The resulting graph need not be identical to a newly constructed graph, so approximate-search quality must be checked separately.

3.2 Global and Region-Local Maps

We first learn a global map from paired examples. We sample a fitting set P_0 of n_0 documents uniformly and encode them with h, obtaining pairs (x_d, y_d). The map $M_0 : \mathbb{R}^{d_0} \to \mathbb{R}^{d_1}$ predicts a new embedding from an old one. We consider the three map families used by Drift-Adapter [31]:

- Orthogonal Procrustes (OP) [26] aligns the spaces through a rotation or reflection. We append zero coordinates when $d_0 < d_1$.
- A low-rank affine map (LA), $M_0(x) = (UV^\top + \mathrm{diag}(s))x + b$, combines a factored linear transformation with a diagonal term and an offset. Here U and V are learned factor matrices, $\top$ denotes transpose, s contains diagonal coefficients, and b is a learned offset vector.
- A residual multilayer perceptron (MLP), $M_0(x) = Wx + b + \mathrm{mlp}(x)$, adds a one-hidden-layer neural network to the linear map with weight matrix W. The residual term learns deviations from that linear prediction.

Each family minimizes $\sum_{d \in P_0} \|y_d - M_0(x_d)\|_2^2$, the total squared prediction error on the fitting pairs. We map documents forward into the new space. This direction allows a newly computed document vector to replace its prediction without changing the query model.

A global map may fit some topics better than others. For example, documents about tower construction and documents about tourism may respond differently to a model upgrade. We group old vectors into R regions using k-means clustering, which assigns vectors to nearby learned centres. In the name k-means, k is the generic cluster count; our cluster count is R, independent of the retrieval cutoff k. When R appears as a subscript below, it identifies a particular region. We then fit an affine correction to the global prediction:

$$\begin{aligned} M_R(x) &= A_R\, M_0(x) + b_R, \\ (A_R, b_R) &= \arg\min_{A,b} \sum_{d \in P_R} \|y_d - AM_0(x_d) - b\|_2^2 + \lambda \|A - I\|_F^2 + \lambda \|b\|_2^2, \end{aligned} \tag{2}$$

Here P_R contains the available fitting pairs in region R. The matrix A_R and offset b_R adjust the global output. The identity matrix I leaves that output unchanged, and $\|\cdot\|_F$ is the Frobenius norm, the square root of the sum of squared matrix entries. The ridge weight λ penalizes departures from $A_R = I$ and $b_R = 0$. Thus the first term fits the observed pairs, while the other terms discourage a large correction from limited data. The operator arg min returns parameters that minimize the stated objective.

We fit this ridge regression using sufficient statistics: summaries that contain the information needed for the fit. They are $\sum_{d \in P_R} u_d u_d^\top$ and $\sum_{d \in P_R} y_d u_d^\top$, where $u_d = (M_0(x_d), 1)$ appends a constant coordinate to handle the offset. A region with fewer than $n_{\min}$ pairs uses the global map. Each re-embedded document supplies another pair, allowing us to update the statistics, refit the regional map, and overwrite the remaining predictions in that region. This creates an opportunity to improve even documents that have not yet been re-embedded. We

retain their old vectors on disk for this purpose and include the extra overwrites in the planned cost accounting.

To assess a region's predictions, we reserve held-out pairs H_R that are not used in its fit. Their root-mean-square vector error is

$$r_R = \Big(\tfrac{1}{|H_R|} \sum_{d \in H_R} \|\mathrm{nrm}(M_R(x_d)) - y_d\|_2^2 \Big)^{1/2} \tag{3}$$

where $|H_R|$ is the number of held-out pairs. The statistic r_R summarizes the typical vector error in the region. It is distinct from the score-error radius r used by the scheduler and does not by itself bound every document's error.

3.3 Score Calibration across Fidelities

The two fidelity classes can have different score distributions. A map trained to minimize squared error may pull predictions towards a regional mean. If mapped vectors then score systematically lower, an equally relevant re-embedded document can receive an advantage simply because its vector was recomputed. We address this possible bias with *calibration*: a scale and offset fitted to align mapped scores with new-model scores.

Let C be a held-out set of query–document pairs. We define the calibrated score $\tilde{s}(q, d)$ using a scale a_{ϕ_d} and offset b_{ϕ_d} for each fidelity class:

$$\begin{aligned} \tilde{s}(q,d) &= a_{\phi_d} \langle y_q, z_d \rangle + b_{\phi_d}, \\ (a_{\mathrm{m}}, b_{\mathrm{m}}) &= \arg\min_{a,b} \sum_{(q,d) \in C} \big(\langle y_q, y_d \rangle - a \langle y_q, z_d \rangle - b \big)^2, \end{aligned} \tag{4}$$

The subscript m denotes the mapped class. Its parameters $(a_{\mathrm{m}}, b_{\mathrm{m}})$ minimize the discrepancy between calibrated mapped scores and directly computed scores. For re-embedded vectors we use $(a_{\mathrm{true}}, b_{\mathrm{true}}) = (1, 0)$, so their scores are unchanged. We form C from sampled logged queries and documents among their highest-scoring mapped results.

We apply the correction when merging search candidates. A positive fitted scale preserves ordering within the mapped class, so calibration can affect the ranking between classes once backfill begins. At zero backfill, such a monotone correction leaves the ranking unchanged. The least-squares expression above does not itself constrain the scale to be positive; this condition matters when interpreting the monotonicity claim.

3.4 Traffic-Driven Backfill

We next decide which document vectors to recompute. The useful candidates are relevant documents close enough to the ranking cutoff that mapping error could displace them. In our example, improving an Eiffel Tower article may help many repeated queries, whereas improving a rarely retrieved article may have little immediate benefit.

For a logged query q, let $\hat{\sigma}_d = \tilde{s}(q, d)$ be the calibrated mapped score. Write $\hat{T}_k(q)$ for its served top-k set and $\hat{\sigma}_{(k+1)}$ for the next-largest score. The *slack* $\hat{\beta}(q, d) = \hat{\sigma}_d - \hat{\sigma}_{(k+1)}$ measures how far a returned document lies above the cutoff. We estimate a score-error radius r from a high quantile of held-out absolute residuals $|\langle y_q, y_d - z_d \rangle|$. A residual is the difference between a direct score and its approximation. This empirical radius guides scheduling; the formal bounds in Section 4 require a bound that holds for every relevant comparison.

We use $\mathrm{Rel}(q)$ for the documents associated with q by clicks. If clicks are unavailable, we use the served top-k as the relevance proxy. The candidate set is

$$A(q) = \{d \in \hat{T}_k(q) \cap \mathrm{Rel}(q) : r < \hat{\beta}(q, d) \leq 2r\}, \tag{5}$$

which selects relevant returned documents in the intermediate slack band $(r, 2r]$. The true-score analysis in Section 4 explains why recomputing one such document can secure a relevant result under a uniform error bound. In practice, the scheduler uses estimated scores and candidate sets, so this interpretation is conditional on their accuracy.

We mark a query as at risk when $A(q)$ is nonempty and no relevant document has slack above $2r$. Let $\mathcal{R}$ be the set of these queries and S a set of documents chosen for backfill. Our coverage objective is

$$F(S) = \sum_{q \in \mathcal{R}} w_q \, \mathbf{1}\{A(q) \cap S \neq \emptyset\}. \tag{6}$$

where $\mathbf{1}\{E\}$ is one if event E holds and zero otherwise. We count a query's weight once when S contains at least one of its candidates. For example, suppose two queries have weights 100 and 10. A document that covers both contributes 110; adding another document that covers only the first query contributes nothing further. These diminishing returns motivate greedy selection.

Algorithm 1 repeatedly selects the document with the largest *marginal gain*, the additional value of F after adding that document. We use lazy evaluation to avoid recomputing every gain after every selection. The max-heap H stores candidates by their last computed gain; we recheck the largest entry before selecting it. We send each batch of B documents to the GPU, overwrite their vectors, refit affected regions, and recompute affected query candidates. These updates change the objective between batches. Section 4 states the guarantee for a fixed objective.

Baseline schedulers and serving. We compare three alternative orders within the same backfill loop. Random order follows the vendor migration recipe. Uncertainty order, adapted from Hot-Refresh [41], prioritizes large disagreement $\|M_0(x_d) - M_{R(d)}(x_d)\|_2$ between the global and regional maps. Re-embed-on-read queues any mapped document returned in a top-k result, without considering query weights or score margins.

At serving time we encode a query with h, search the mixed index, calibrate the candidate scores using Eq. (4), and return the top-k. We log the query and

Algorithm 1 Traffic-greedy backfill with lazy evaluation

compute $A(q)$ for every logged q by one search on the mapped index, Eq. (5); $S \leftarrow \emptyset$
gain$(d) \leftarrow \sum_{q:d\in A(q)} w_q$ for every candidate d; push all into max-heap H
while $|S| < fN$ **do**
    batch $\leftarrow \emptyset$; **repeat** pop d with the largest stored gain, gain$(d) \leftarrow F(S \cup \text{batch} \cup \{d\}) - F(S \cup \text{batch})$,
    add d to batch if gain$(d) \geq$ the largest stored gain in H, else push it back **until** $|\text{batch}| = B$ or $H = \emptyset$
    re-embed batch; overwrite $z_d \leftarrow y_d$, $\phi_d \leftarrow$ true; $S \leftarrow S \cup$ batch; refit touched M_R and overwrite their mapped vectors; recompute affected $A(q)$, update H
end while

the fidelity flags of its results. The *staleness fraction* is the fraction of returned documents whose vectors remain mapped.

4 Guarantees

We separate three questions: when mapping preserves the exact ranking, when backfill preserves at least one relevant result, and how well greedy selection optimizes the coverage objective. The ranking arguments compare scores over all documents. Approximate HNSW traversal can introduce additional search errors, which these arguments do not bound.

Intuition: the gap must exceed the uncertainty. Suppose the two highest direct scores for a query are 0.80 and 0.70. If each mapped score differs from its direct score by at most 0.03, the first can fall to 0.77 and the second rise to 0.73. Their order is preserved because the 0.10 gap exceeds twice the error bound. With scores 0.80 and 0.75, that argument fails. These are illustrative scores, not measurements. Theorem A extends this comparison to the boundary between the k-th and $(k+1)$-th documents.

Fix a query q and unit-length vectors. Let $\sigma_d = \langle y_q, y_d \rangle$ be document d's direct new-model score, and let $\sigma_{(j)}$ denote the j-th largest such score. The full-rebuild top-k set is $T_k(q)$, with ties broken by a fixed order. Its *margin* is $\gamma_k(q) = \sigma_{(k)} - \sigma_{(k+1)}$.

For this score analysis, we write $\hat{\sigma}_d = \langle y_q, z_d \rangle$ for the raw stored-vector score and $\hat{T}_k(q)$ for its exact top-k set. This differs from the calibrated score denoted by $\hat{\sigma}_d$ in the operational scheduler. The *score residual* $\epsilon_d = \sigma_d - \hat{\sigma}_d = \langle y_q, y_d - z_d \rangle$ measures their difference. Assume a per-document bound $|\epsilon_d| \leq \rho_d$, where $\rho_d = 0$ for a re-embedded document. Applying the analysis to calibrated scores requires bounds on those calibrated residuals.

Theorem A (Margin lemma). *If* $\gamma_k(q) > \max_{d\in T_k(q)} \rho_d + \max_{d'\notin T_k(q)} \rho_{d'}$ *then* $\hat{T}_k(q) = T_k(q)$. *In particular, if every mapped document satisfies* $|\epsilon_d| \leq r$, *then* $\gamma_k(q) > 2r$ *suffices, and* $\gamma_k(q) > r$ *suffices when all of* $T_k(q)$ *or all documents outside it are re-embedded.*

Proof. For $d \in T_k(q)$ and $d' \notin T_k(q)$, $\hat{\sigma}_d \geq \sigma_{(k)} - \rho_d$ and $\hat{\sigma}_{d'} \leq \sigma_{(k+1)} + \rho_{d'}$, hence $\hat{\sigma}_d - \hat{\sigma}_{d'} \geq \gamma_k(q) - \rho_d - \rho_{d'} > 0$. Every member of $T_k(q)$ outscores every non-member, so the k largest stored scores are exactly $T_k(q)$.

The special cases substitute r or 0 for the maxima. □

Vector residuals and score residuals. The Cauchy–Schwarz inequality gives $|\epsilon_d| \leq \|y_d - z_d\|_2$: a unit query cannot amplify the length of a vector error. Thus a bound on an individual vector residual also bounds its score residual. The regional statistic r_R in Eq. (3) averages squared errors; it is not an individual worst-case bound.

Maystre et al. [16] analyze orthogonal alignment. If pairwise inner products of old and new embeddings differ by δ in root mean square, their Procrustes bound is $\mathbb{E}_d\|M_0 x_d - y_d\|_2^2 \leq \sqrt{2d_1}\,\delta$ after zero-padding. Here $\mathbb{E}_d$ is expectation over documents, and δ measures the change in their pairwise geometry. Markov's inequality turns this average squared-error bound into a bound on the fraction of documents exceeding a chosen radius ρ. Normalization can at most double the vector error under the stated unit-target setting.

The vector bound can be conservative in high dimension. Heuristically, an error with no preferred direction relative to the query has a projection near $\|y_d - z_d\|_2/\sqrt{d_1}$. This resembles the random-projection intuition behind the Johnson–Lindenstrauss lemma [12]; it is not an additional guarantee for our learned maps. Calibration targets systematic score bias. The scheduler estimates r from a high residual quantile on held-out pairs near the top-k cutoff, whereas the following corollary assumes a radius valid for every mapped document.

From exact rankings to relevant hits. A search can remain useful even if the complete ranking changes: it may still return a relevant document. For $d \in T_k(q)$, define its true-score slack as $\beta(q, d) = \sigma_d - \sigma_{(k+1)}$. Let $\mathrm{Rel}(q)$ be the relevant set and $\mathrm{hit}_k(q; S) = \mathbf{1}\{\mathrm{Rel}(q) \cap \hat{T}_k(q) \neq \emptyset\}$ after re-embedding the documents in S. This indicator records whether at least one relevant document is returned.

We restrict attention to queries for which the full rebuild returns a relevant document strictly within the top-k, with no tie at the cutoff. Let w be the traffic distribution normalized over those queries; $w(\text{safe})$, for example, is the total probability mass of safe queries. We divide queries into three groups:

- A query is *safe* if some relevant $d^* \in T_k(q)$ has $\beta(q, d^*) > 2r$. Even errors on both sides of the cutoff cannot displace that document.
- It is *rescuable* if it is not safe and some relevant d^* has $r < \beta(q, d^*) \leq 2r$. Re-embedding that document removes its own score error.
- It is *hard* otherwise: this sufficient-condition argument cannot certify a relevant hit through one such replacement.

For the corollary, $A(q)$ and F are the true-score counterparts of Eqs. (5) and (6), with normalized weights. The operational scheduler estimates these sets from mapped scores.

Corollary B (Recall of the mixed index). *Assume $|\epsilon_d| \le r$ for every mapped document. A relevant $d^* \in T_k(q)$ is served whenever $\beta(q, d^*) > r + \rho_{d^*}$. Hence every safe query is answered, a rescuable query is answered as soon as $A(q) \cap S \neq \emptyset$, and*

$$\mathbb{E}_{q\sim w}\big[\mathrm{hit}_k(q; S)\big] \;\ge\; w(\mathit{safe}) + F(S) \;\ge\; 1 - w(\mathit{rescuable}) - w(\mathit{hard}).$$

Proof. d^* leaves $\hat{T}_k(q)$ only if k documents have stored score at least $\hat{\sigma}_{d^*}$, and since $T_k(q)\setminus\{d^*\}$ has $k-1$ elements, some $d' \notin T_k(q)$ must satisfy $\hat{\sigma}_{d'} \ge \hat{\sigma}_{d^*}$. But $\hat{\sigma}_{d'} \le \sigma_{(k+1)} + r$ and $\hat{\sigma}_{d^*} \ge \sigma_{d^*} - \rho_{d^*}$, which is impossible when $\beta(q, d^*) > r + \rho_{d^*}$.

With $\rho_{d^*} \le r$ the safe case follows and with $\rho_{d^*} = 0$ for $d^* \in S$ the rescuable case.

Summing w_q over answered queries gives the first inequality, and at $S = \emptyset$ the answered queries include every safe query. □

The corollary gives a lower bound on the traffic-weighted hit rate, not on every document's recall. Its condition is sufficient: a query classified as hard may still return relevant documents. The bound is conservative because it considers the worst permitted score change for every competitor.

The scheduler estimates slack from stored scores. If every score error is bounded by r, the $(k+1)$-th ordered score changes by at most r, and a slack estimate can differ from its true value by at most $2r$. Thus estimated and true bands need not coincide. Section 6 describes how we plan to test whether the estimated bands are useful.

Why greedy selection has a bound. For a fixed collection of candidate sets, selecting another document never removes coverage: F is *monotone*. A document's additional value can only decrease as other documents cover the same queries: F is *submodular*, the formal diminishing-returns property. Greedy selection repeatedly chooses the largest remaining gain. In Theorem C, m is the number of documents allowed by the budget, S_m is the resulting set, and e is the base of the natural logarithm. The guarantee compares its coverage with the best set of at most m documents.

Theorem C (Traffic-greedy is a $(1-1/e)$-approximation). *F is monotone and submodular. For a budget of m documents the greedy schedule S_m, which adds the document of largest marginal gain at each step, satisfies $F(S_m) \ge (1 - 1/e)\max_{|S|\le m} F(S)$. For a fixed F, lazy evaluation returns the same schedule and releasing B consecutive picks as one batch does not change the sequence of picks.*

Proof. For $S \subseteq T$ and $d \notin T$, $F(S\cup\{d\}) - F(S) = \sum_{q\in\mathcal{R}} w_q \mathbf{1}\{d \in A(q),\ A(q)\cap S = \emptyset\}$. Since $A(q)\cap T = \emptyset$ implies $A(q)\cap S = \emptyset$ and weights are non-negative, every term of the gain at T is a term of the gain at S (submodularity), and $F(S) \le F(T)$ (monotonicity).

Let O be optimal with $|O| \le m$, S_i the greedy set after i steps and $\Delta_d(S) = F(S\cup\{d\}) - F(S)$. Monotonicity, submodularity and the greedy choice give

$F(O) \le F(S_i \cup O) \le F(S_i) + \sum_{d \in O \setminus S_i} \Delta_d(S_i) \le F(S_i) + m\,(F(S_{i+1}) - F(S_i))$, hence $F(O) - F(S_{i+1}) \le (1 - 1/m)\,(F(O) - F(S_i))$ and, iterating from $S_0 = \emptyset$, $F(O) - F(S_m) \le (1 - 1/m)^m F(O) \le e^{-1} F(O)$ [21].

Submodularity makes every stored marginal gain an upper bound on the current one, hence the largest re-evaluated gain that still dominates the heap is the true maximum, the lazy-greedy argument of Minoux [18]. □

Theorem C applies to a fixed F. Algorithm 1 refits maps and recomputes $A(q)$ after each batch, so it changes F between batches. The fixed-objective argument therefore applies within a batch epoch, with the candidate sets frozen at its start; it does not establish a single approximation ratio for the entire adaptive migration.

If documents have different re-embedding costs, such as different token lengths, the budget becomes a *knapsack constraint*: the sum of selected costs must fit the budget. The gain-per-token greedy variant, compared with the best single document, gives $(1 - 1/e)/2$ [13]. Enumerating small initial sets before greedy completion can restore $(1 - 1/e)$ [29]. These are distinct cost models and algorithm variants.

Why a map alone cannot converge. Mapping can reuse information already encoded in old vectors, but it cannot recover document distinctions that those vectors have lost. The basic linear identity is $\langle y, M x_d \rangle = \langle M^\top y, x_d \rangle$, where y is a new-space query and $M^\top$ is the transpose of M. Before document-dependent normalization, scoring mapped vectors is therefore equivalent to scoring old vectors with the transformed query $M^\top y \in \mathbb{R}^{d_0}$.

Weller et al. [36] bound the number of top-k sets representable in a given dimension through the *sign rank* of a relevance matrix: the minimum rank needed to reproduce its pattern of positive and negative entries. This connects representational limits to retrieval. Regional maps and the nonlinear residual MLP can express more than one global linear map, but their document inputs are still old embeddings. Re-embedding supplies information directly from the original text. The linear identity alone does not characterize the complete normalized, calibrated mixed index.

5 Experimental Setup

This section specifies the planned evaluation. The artifact has not yet run, and the numerical outputs in Section 6 are placeholders. We describe configurations and comparisons as a protocol to be executed.

Datasets and traffic. We plan to use five BEIR collections [30] also considered by closely related work [8,16]: Natural Questions (NQ; 2.68 M documents, 3,452 queries), HotpotQA (5.23 M, 7,405), FEVER (5.42 M, 6,666), TREC-COVID (171 k, 50), and FiQA (57.6 k, 648). Here M and k abbreviate million and thousand. These collections provide relevance judgments for ranking evaluation. We

use normalized discounted cumulative gain at rank 10 (nDCG@10), which rewards relevant results near the top and normalizes by the best possible ordering under the judgments.

The scale study uses MS MARCO documents [2] with 5,193 judged development queries. ORCAS [5] provides the traffic replay: 10 M distinct queries and 18 M clicked query–document pairs covering 1.4 M documents in the draft configuration. These macro-supplied counts await verification. Because ORCAS has no timestamps, we will split query strings randomly into scheduling and held-out halves. The scheduler will not see the held-out half. We will also remove strings overlapping the judged development set; 2,073 is the current placeholder for that count.

For temporal replay, we plan to use AOL-IA, the AOL search log with documents recovered from the Internet Archive [14]. The draft contains 1.5 M documents and 36 M timestamped entries, divided into 8 weekly slices. At week t, the scheduler can read weeks 1 through t; we evaluate on week $t + 1$. Clicked documents define relevance for scheduling only. Evaluation uses judged development queries or held-out traffic compared with the full rebuild, so the scheduler is not evaluated on its own optimization queries.

Model pairs. We specify four upgrades. The nomic-embed-text v1 to v1.5 pair [22] is a same-family refresh with 768-dimensional vectors. The snowflake-arctic-embed-l to l-v2.0 pair [17,40] is specified at 1024 dimensions with a new XLM-R encoder backbone; it is our planned scale, ablation, theory, and cost case. The bge-base-en-v1.5 to bge-m3 pair [37,4] moves from 768 to 1024 dimensions. Finally, all-MiniLM-L6-v2 to all-mpnet-base-v2 moves from 384 to 768 dimensions and matches the pair studied by Drift-Adapter [31]. Checkpoint dimensions remain subject to verification before execution.

The full new-model embeddings provide the rebuild reference. The migration procedure may access new embeddings only for the n_0 fitting documents and the backfill batches it requests. This distinction prevents reference information from leaking into scheduling.

Comparison systems. The *full rebuild* is the reference system with every document re-embedded. The *misaligned* baseline searches old document vectors with new query vectors when their dimensions agree. We plan query-side implementations of Drift-Adapter's orthogonal Procrustes (OP), low-rank affine (LA), and residual-MLP maps. They use the same n_0 paired documents as our maps, plus a 20k-pair configuration matching the published setup. The planned LA rank is 64; the MLP has 256 hidden units and is trained for 50 epochs with Adam, an adaptive gradient optimizer.

Query drift compensation (QDC) [8] subtracts the mean drift vector. minivec2vec [6] fits an orthogonal map using 60k unpaired vectors from each space. Relative representations [19] compare cosine similarities to 1,024 anchor documents encoded by both models. These baselines test progressively different ways of aligning representations.

The *dual index with random backfill* follows the migration pattern documented by Qdrant [23]. We search the old index with the old query model and the new index with the new model, then rerank the combined candidates using new-model scores, approximating old-only candidates through a Drift-Adapter map. We also compare uncertainty order and re-embed-on-read from Section 3.4. A *hindsight oracle* uses information unavailable online: it selects, by ORCAS weight, documents whose replacement actually changes a development query's result. This supplies an informative comparison rather than a deployable scheduler.

Our configuration. We specify a residual-MLP global map, $R = 256$ regions, ridge weight $\lambda = 10$, and fallback to the global map below $n_{\min} = 20$ pairs. Calibration uses 2,000 held-out query–document pairs near the logged queries' top-k cutoff. The scheduler estimates r as the 95th percentile of their absolute score residuals and processes batches of 1% of the corpus. MixIndex-G denotes the global map alone; MixIndex-R denotes the full system.

The planned index implementation is hnswlib with $M = 16$, $ef_{\text{construction}} = 200$, and $ef_{\text{search}} = 100$, returning $k = 10$ results. Here M controls graph connectivity, while the two ef parameters control the candidate-list sizes during construction and search. A layer around the index stores fidelity flags and performs overwrites.

Metrics. We use several complementary measures:

- *ARR* is Recall@10 against the full rebuild: the fraction of its ten highest-scoring documents recovered in the served top ten. It measures agreement with the reference ranking.
- *nDCG@10 retained* divides judged nDCG@10 by the full rebuild's value and expresses the ratio as a percentage. *Hit@10 retained* uses the same ratio for the fraction of queries retrieving at least one relevant document.
- *NFR@1*, the negative flip rate [41], measures queries for which the full rebuild's top result was relevant but the migrating system's top result is not.
- The *time-to-quality curve* plots retained nDCG@10 against the fraction re-embedded, checked after each batch. Its normalized integral summarizes average retained quality over the migration budget. The *staleness fraction* is the share of returned top-ten vectors still mapped.

AOL-IA has no relevance judgments, so its quality measure is traffic-weighted ARR against the full rebuild. Agreement with a reference ranking should not be confused with independent evidence of relevance.

Cost and uncertainty. We will record GPU-hours, index construction and overwrite time, downtime, and query latency under 16 client threads at 11,800 queries per second. A GPU-hour is one GPU in use for one hour. The latency percentiles p50 and p99 are the median and 99th percentile of response times. We will apply overwrites as batches arrive to measure serving under concurrent migration.

Map fitting uses 3 random seeds and traffic experiments use 5. We will report means and 95% Student-t confidence intervals across seeds; with few seeds, these intervals can be wide. The current template reports the largest half-width. A paired bootstrap will resample queries jointly across compared systems, with Holm correction to account for multiple comparisons across model pairs. No confidence interval or significance result has yet been measured.

Hardware. The planned embedding device is one NVIDIA H100 with 80 GB of memory. Map fitting, index construction, overwrites, scheduling, replay, and latency evaluation are assigned to an Apple M5 Max workstation with 68 GB of unified memory. These are evaluation settings, not evidence that the artifact has completed a run.

6 Evaluation Questions and Illustrative Results

Status of this section. Every numerical outcome, table, and curve in this section is an **illustrative placeholder**. The values were chosen to populate the reporting template; no experiment, confidence interval, or significance test has been run. We retain the generated assets so that measured outputs can replace them. Below, we explain what each comparison is intended to test and how to read its fields. The example values do not establish an ordering of methods or support causal explanations.

Q1: quality before backfill. We first ask whether mapping provides a useful starting point for migration. "Zero backfill" means no scheduled corpus backfill after fitting; fitting pairs still require new-model embeddings. Table 1 compares map-based methods averaged over the five BEIR collections. Its illustrative misaligned values are 75.4% nDCG@10 retained for the nomic pair and 40.2% for arctic. The QDC field for arctic is 47.5%. These comparisons are designed to distinguish the cost of an uncorrected space mismatch from the benefit of mapping.

The template includes Drift-Adapter ARR values from 0.951 to 0.975 on MiniLM to mpnet, with 0.977 for the MLP fitted on 20k pairs. These placeholders must not be treated as a reproduction of its published 0.95–0.99 range. We will compare the query-side MLP with our forward global map, MixIndex-G, to assess the effect of mapping direction. Their illustrative mean nDCG@10 values are 96.4% and 96.0%, respectively.

For MixIndex-R, the current fields are 97.2% nDCG@10 retained and ARR 0.970, compared with 0.962 for Drift-Adapter MLP. The arctic per-collection range is 95.1% to 98.6%. The bge comparison is 95.5% against 94.3%, while mini-vec2vec and relative representations have mean fields of 91.6% and 88.7%. Measured results will determine whether any benefit is consistent across upgrades and collections.

The statistical reporting fields are a largest 95% confidence half-width of 0.3 points and a comparison threshold $p < 0.01$. They are placeholders as well. No statistical rejection or uncertainty estimate can be inferred from these values.

Table 1. Illustrative placeholders; experiments not run. Zero-backfill quality on the BEIR average (five sets), mean over 3 seeds. ARR is Recall@10 agreement with the full rebuild, nDCG is nDCG@10 retained in percent, DA is Drift-Adapter, and a dash marks a method that cannot run across dimensions.

	nomic v1→1.5		arctic-l→v2		bge-base→m3		MiniLM→mpnet	
Method	ARR	nDCG	ARR	nDCG	ARR	nDCG	ARR	nDCG
Full rebuild (abs.)	–	0.548	–	0.566	–	0.581	–	0.469
Misaligned	0.712	75.4	0.318	40.2	–	–	–	–
DA OP	0.962	96.5	0.931	93.6	0.903	91.2	0.951	95.7
DA LA	0.971	97.2	0.944	94.8	0.922	92.9	0.963	96.6
DA MLP	0.979	97.8	0.957	95.9	0.938	94.3	0.975	97.5
QDC	0.834	86.0	0.402	47.5	–	–	–	–
mini-vec2vec	0.941	94.7	0.902	91.0	0.868	88.0	0.921	92.8
Rel. repr.	0.903	91.6	0.861	87.4	0.842	85.5	0.889	90.1
MixIndex-G	0.975	97.5	0.952	95.5	0.933	93.9	0.972	97.2
MixIndex-R	0.983	98.4	0.968	97.0	0.951	95.5	0.980	98.1

Q2: quality as migration progresses. Figure 2 and Table 2 specify the arctic migration comparison on MS MARCO. We will evaluate quality after each backfill batch, using the same initial index for the schedulers. The illustrative starting value is 95.6%; the fixed query-side map has a flat field of 94.7%. Re-embedding all documents eventually supplies the full set of new-model vectors, although graph-search behavior must still be checked separately.

At 10% backfill, the template contains 96.0% for random order, 97.0% for uncertainty order, 97.4% for re-embed-on-read, and 98.6% for traffic-greedy. At 25%, the traffic-greedy field is 99.2%. These entries will test whether query-weighted selection yields useful quality earlier in migration. The normalized curve integrals, currently 0.994 and 0.978 for greedy and random order, summarize performance across the full budget rather than at one checkpoint.

We also account for work performed before backfill begins. The illustrative 10% checkpoint costs 0.20 GPU-hours, including fitting-pair embeddings, compared with 1.9 for the full corpus. The saving field is 89%. Measured cost must include all fitting and migration work before such a saving can be claimed.

To test generalization beyond the scheduling log, the held-out ORCAS template records ARR 0.990 for greedy and 0.961 for random order at 10%, starting from 0.948. The AOL-IA temporal replay records integrals of 0.989, 0.984, 0.983, and 0.979 for greedy, uncertainty, re-embed-on-read, and random order. This replay will test whether documents selected from earlier weeks remain useful in later traffic.

Q3: contributions of maps and calibration. An *ablation* removes one component while keeping the others fixed. Table 3 uses this comparison to separate regional mapping from calibration. Its zero-backfill arctic fields are 94.9% for the global MLP and 95.6% with regional maps. Figure 3 (right) specifies the corresponding vector-error comparison: mean residuals 0.257 and 0.211, and worst-decile fields 0.41 and 0.30. A decile groups one tenth of the regions. These measurements will test whether local corrections reduce errors concentrated in particular regions.

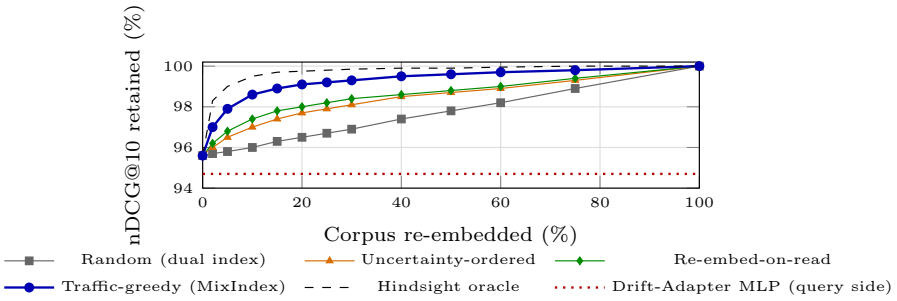


Fig. 2. Illustrative placeholders for the time-to-quality comparison on MS MARCO with the arctic pair. The horizontal axis is the share of documents re-embedded; the vertical axis is retained nDCG@10 on held-out development queries. The planned aggregate is a mean over 5 random seeds. These curves are not measured results.

Table 2. Illustrative placeholders; experiments not run. Backfill schedulers on MS MARCO documents, arctic pair, mean over 5 seeds: nDCG@10 retained (%) after re-embedding 5, 10, 25 and 50% of the corpus, the integral of the curve over the full budget, and traffic-weighted ARR on the held-out ORCAS window at 10%.

Scheduler	5%	10%	25%	50%	Integral	$ARR_{10\%}$
Random (dual index)	95.8	96.0	96.7	97.8	0.978	0.961
Uncertainty-ordered	96.5	97.0	97.9	98.7	0.985	0.970
Re-embed-on-read	96.8	97.4	98.2	98.8	0.987	0.982
Traffic-greedy (MixIndex)	97.9	98.6	99.2	99.6	0.994	0.990
Hindsight oracle	99.0	99.5	99.8	99.9	0.998	0.994

Calibration is intended to matter when mapped and re-embedded vectors coexist. At 10% backfill, the illustrative global-map fields are 97.6% without calibration and 98.0% with it; the full-system field is 98.6%. The NFR@1 fields are 3.3% and 2.3%. The score-bias comparison currently contains 0.031 for the global map, 0.019 for regional maps, and 0.002 after calibration. These values are reporting examples rather than evidence that calibration removes bias.

We will also compare the proportion of mapped vectors in the corpus with their proportion in returned results. Under random 10% backfill, the illustrative population fraction is 0.90, while the returned fractions are 0.81 without calibration and 0.89 with it. This comparison can reveal a representation-type imbalance, although relevance and document selection must be considered before attributing a difference to calibration.

Finally, we vary configuration choices. The region-count fields are 95.3% for $R = 64$ and 95.6% for $R = 256$. Fitting sets of 2k, 5k, 10k, and 20k pairs have fields 93.8, 95.0, 95.6, and 95.7%. These sweeps will test sensitivity to training data and regional granularity, including the saturation behavior discussed by Maystre et al. [16].

Table 3. Illustrative placeholders; experiments not run. Ablation on MS MARCO documents at zero backfill and after re-embedding 10% with the traffic-greedy scheduler (nDCG@10 retained, %), and NFR@1 (%) at 10% for the arctic pair. At 0% every vector is mapped and calibration cannot change the ranking.

	arctic-l → l-v2.0		bge-base → bge-m3		NFR@1
Variant	0%	10%	0%	10%	10%
Global map, OP	93.0	96.4	90.4	94.6	4.3
Global map, LA	94.1	97.0	92.0	95.5	3.7
Global map, MLP	94.9	97.6	93.2	96.4	3.3
+ regional maps	95.6	98.1	94.4	97.0	2.9
+ calibration	94.9	98.0	93.2	96.9	2.7
+ both (MixIndex)	95.6	98.6	94.4	97.6	2.3

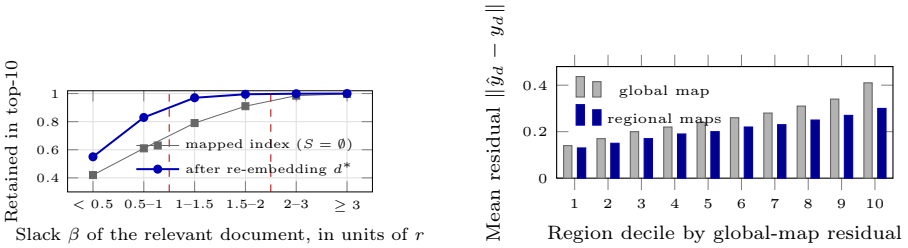


Fig. 3. Illustrative placeholders for the theory and mapping checks. Left: whether a relevant document remains in the top ten, grouped by its score slack relative to radius r. Right: mean vector error for groups of regions ordered by residual size. The bands refer to Corollary B's sufficient conditions; the plotted values have not been measured.

Q4: margin conditions and scheduling coverage. The theory check asks whether the estimated slack bands help identify useful backfill candidates. The template uses $r = 0.014$, with 0.018 before calibration, and traffic-weighted shares of 62.4% safe, 20.9% rescuable, and 16.7% hard queries. Figure 3 (left) groups relevant full-rebuild results by their slack. The illustrative retention fields run from 0.42 in the lowest band to 0.985 above $2r$. Re-embedding a candidate has a field of 0.97 in the intermediate band and a rescue-rate field of 94.2% over rescuable pairs.

We will compare a conditional certified hit-rate bound with actual hit rate. The current bound fields are 62.4% before backfill and 80.8% after covering 88% of rescuable traffic; the corresponding empirical-result fields are 96.1% and 98.9%. Every field is illustrative. In particular, an estimated quantile radius does not turn this comparison into a guarantee for all queries.

For the fixed coverage objective, we will compare greedy selection with a linear-programming relaxation, an optimization that permits fractional selections and yields an upper bound on the discrete optimum. The template records 97.4% of that bound at a 10% budget. It also records 3.1% as the fraction of gain evaluations needed by lazy versus ordinary greedy selection. These checks con-

Table 4. Illustrative placeholders; experiments not run. Migration cost on MS MARCO documents (3.2 M vectors, arctic pair, HNSW $M = 16$, $ef = 200$): GPU-hours on one H100, HNSW build (rebuild, dual) or map fitting plus every overwrite of the budget, refits included (MixIndex), on the M5 Max, latency during the migration, in-memory index copies (old vectors stay on disk), nDCG@10 retained at first serve and at the end of the budget.

Strategy	GPU-h	Index min	p50 ms	p99 ms	Down s	Mem.	Q_{first}	Q_{end}
Full rebuild	1.90	38.0	0.9	2.4	41	2.0×	93.5	100.0
Dual index, random	1.90	38.0	1.6	4.1	0	2.0×	93.5	100.0
MixIndex, 10% budget	0.20	3.1	1.0	2.9	0	1.0×	95.6	98.6
MixIndex, 25% budget	0.48	4.9	1.0	2.9	0	1.0×	95.6	99.2

cern objective optimization and computational work, separately from retrieval quality.

Systems cost and approximate search. Table 4 retains the cost reporting template. Full re-embedding has fields of 1.9 GPU-hours at 470 documents per second, with 38 minutes for index construction. The old-model quality field during rebuilding is 93.5%, and the cutover field is 41 seconds. The dual-index and baseline p99 latency fields are 4.1 and 2.4 milliseconds.

For MixIndex, the global-map, regional-map, and calibration fitting fields are 71, 24, and 3 seconds. Computing the scheduling objective has a field of 6.4 CPU minutes, followed by 48 seconds for greedy selection at 10% backfill. We will count additional regional overwrites explicitly: the illustrative amplification is 1.6 overwritten vectors per re-embedded document. Applying the 10% budget's overwrites with neighbour repair has a field of 88 seconds, with p50 and p99 latency fields of 1.0 and 2.9 milliseconds.

The template's index time combines fitting and overwrites. Its quality–cost checkpoints pair 98.6% with 0.20 GPU-hours and 99.2% with 0.48 GPU-hours. After all vectors have been overwritten, we will compare Recall@10 against exact search for repaired, unrepaired, and freshly built graphs. Their current fields are 0.987, 0.979, and 0.988 at $ef_{\text{search}} = 100$. This planned check evaluates the overwrite primitive studied by Embedding Surgery [1]; the placeholders do not confirm its behavior on our corpus.

7 Discussion and Limitations

When gradual migration could help. Our design is motivated by collections in which a small fraction of documents accounts for much of the traffic. The planned ORCAS replay covers 1.4 M clicked documents within a corpus of 3.2 M documents; these draft values remain placeholders. Concentrated traffic could let a limited backfill budget improve frequently used results early. If traffic is nearly uniform, weighting by query frequency offers less discrimination. If mapping errors are large relative to score margins, substantial re-embedding may still be

necessary. The placeholder curves do not establish either benefit or failure mode empirically.

Relevance and workload shift. Clicks are our relevance proxy for logged queries. They reflect what the old search system exposed, so they can favor already well-ranked documents. We plan held-out ORCAS queries and later AOL-IA weeks to evaluate generalization, although these splits cannot remove all click bias. Without clicks, the scheduler falls back to the served top-k; this fallback needs a separate evaluation.

What the analysis covers. The margin and hit-rate results require valid score-error bounds. A radius estimated as a high quantile on held-out pairs does not bound every document. Approximate graph search and score calibration introduce further distinctions from the exact raw-score analysis. The greedy guarantee concerns fixed candidate sets; refitting maps changes the objective between batches.

Implementation scope. Our proposed evaluation uses HNSW. Embedding Surgery [1] also studies overwrites in DiskANN, but our artifact has not established results for that index. A quantized index, which stores compressed vector codes, would need mapped vectors to be quantized again. These implementation and quality questions remain open until the experiments are run.

8 Conclusion

We propose MixIndex to make an embedding-model upgrade a gradual data migration. We first map old document vectors into the new space, then replace selected predictions with vectors computed directly from the text. Regional maps use the resulting training pairs, score calibration addresses differences between representation types, and the scheduler uses query traffic to choose backfill work. Our analysis explains how score margins constrain ranking errors and why greedy selection approximates a fixed coverage objective. The planned evaluation will test whether these mechanisms improve quality early in migration while keeping one serving index. The current numerical results are placeholders, so the empirical quality and cost benefits remain to be established.

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